



BAULKHAM HILLS HIGH SCHOOL

2017

**YEAR 12 HALF YEARLY
HIGHER SCHOOL CERTIFICATE
EXAMINATION**

Mathematics Extension 2

General Instructions

- Reading time – 5 minutes
- Working time – 2 hours
- Write using black or blue pen
- Black pen is preferred
- Board-approved calculators may be used
- A reference sheet is provided at the back of this paper
- All relevant mathematical reasoning and/or calculations must be shown

Total marks – 70

Section I (Pages 2-4)

10 marks

Attempt Questions 1-10

Allow about 15 minutes for this section

Section II (Pages 5-9)

60 marks

Attempt Questions 11-14

Allow about 1 hours 45 minutes for this section

Section I

10 marks

Attempt Questions 1-10

Allow about 15 minutes for this section

Use the multiple-choice answer sheet for Questions 1 to 10.

- 1 For the ellipse with the equation $\frac{x^2}{16} + \frac{y^2}{25} = 1$. What is the eccentricity of the ellipse?
- (A) $\frac{9}{25}$
(B) $\frac{3}{5}$
(C) $\frac{1}{5}$
(D) $\frac{1}{\sqrt{5}}$
- 2 If $w = \sqrt{3}(\cos \pi + i \sin \pi)$, then w^4 is equal to:
- (A) -9
(B) 9
(C) 81
(D) -81
- 3 The polynomial $P(z) = z^3 + (1 + i)z^2 + (1 + i)z + 1$ has a real zero $z = -1$ and a complex zero $z = \alpha$. The third root is:
- (A) $\frac{1}{\alpha}$
(B) $\bar{\alpha}$
(C) $-\alpha$
(D) $1 - \alpha$
- 4 What is the multiplicity of the root $x = 1$ of the equation $3x^5 - 5x^4 + 5x - 3 = 0$?
- (A) 1
(B) 2
(C) 3
(D) 4

5 z and w are two complex numbers. Which of the following statements is ALWAYS TRUE?

- (A) $|z + w| \geq |z - w|$
- (B) $|z + w| \leq |z - w|$
- (C) $|z| - |w| \geq |z + w|$
- (D) $|z| + |w| \geq |z - w|$

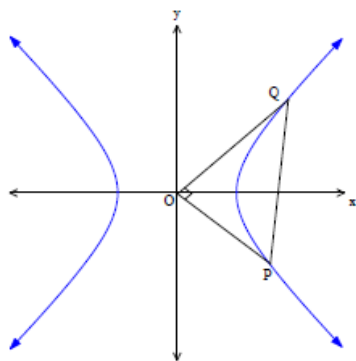
6 The polynomial $P(x) = x^4 + ax^3 + bx^2 + cx + d$ has real coefficients and $P(2i) = P(2 + i) = 0$.

What is the value of $a + b + c + d$?

- (A) 0
- (B) 6
- (C) 9
- (D) 49

7 The diagram below shows the hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ where $a > b > 0$.

The points $P(a \sec \theta, b \tan \theta)$ and $Q(a \sec \alpha, b \tan \alpha)$ lie on the hyperbola and the chord PQ subtends a right angle at the origin.



Which of the following is true?

- (A) $\tan \theta \tan \alpha = -\frac{a^2}{b^2}$
- (B) $\tan \theta \tan \alpha = \frac{a^2}{b^2}$
- (C) $\sin \theta \sin \alpha = -\frac{a^2}{b^2}$
- (D) $\sin \theta \sin \alpha = \frac{a^2}{b^2}$

8 If $x^2 + 3xy = 5y^2$, which of the following is an expression for $\frac{dy}{dx}$?

(A) $\frac{2x+3y}{10y-3x}$

(B) $\frac{2x}{10y-9xy}$

(C) $\frac{2x+3y}{\sqrt{5(x^2+3xy)}}$

(D) $\frac{2x+3y}{10y}$

9 Seven travellers arrive in a town where there are 4 hotels. If two of the travellers want to stay in the same hotel, how many different accommodation arrangements are there?

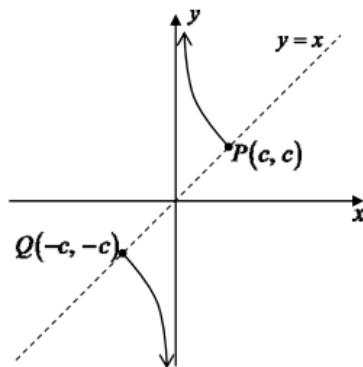
(A) $\frac{6!}{2!2!} \times 2$

(B) $\frac{7!}{2!2!2!} \times 2$

(C) $4^6 \times 2$

(D) 4^6

10 The graph shows a part of the hyperbola $xy = c^2$



Which pair of parametric equations precisely describes the sections of the hyperbola shown?

(A) $x = c(t^2 + 1), y = \frac{c}{t^2+1}$

(B) $x = c(1 - t^2), y = \frac{c}{1-t^2}$

(C) $x = c\sqrt{1 - t^2}, y = \frac{c}{\sqrt{1-t^2}}$

(D) $x = c \sin t, y = \frac{c}{\sin t}$

End of Section I

Section II

90 marks

Attempt questions 11 -14

Allow about 1 hours 45 minutes for this section

Answer each question on the appropriate page of your answer booklet

In Questions 11–14, your responses should include relevant mathematical reasoning and/or calculations.

Question 11 (15 marks)		Marks
(a)	Let $z = 2 + i$ and $w = i - 1$. Find, in the form $x + iy$:	
	(i) $3z + iw$	1
	(ii) $z\bar{w}$	2
(b)	(i) Express $\sqrt{3} + i$ in modulus argument form.	1
	(ii) If $z = \sqrt{3} + i$, hence show that $z^7 + 64z = 0$.	2
(c)	Consider the hyperbola with the equation $\frac{x^2}{16} - \frac{y^2}{9} = 1$.	
	(i) Find the eccentricity of the hyperbola.	1
	(ii) Find the coordinates of the foci and the x -intercepts.	2
	(iii) Find the equations of the directrices and the equations of the asymptotes.	2
(d)	Sketch the region in the complex plane where the inequalities $ z + \bar{z} \leq 1$ and $ z - i \leq 1$ hold simultaneously.	2
(e)	Let ω be one of the complex roots of the equation $z^3 - 64 = 0$.	
	(i) Show that $\omega^2 = -4(\omega + 4)$.	1
	(ii) Hence evaluate $(\omega + 4)^3$.	1
End of Question 11		

Question 12 (15 marks)**Marks**

(a) The roots of $x^3 + 6x^2 + 5x - 2 = 0$ are α, β and γ .

(i) Find the monic polynomial with roots α^2, β^2 and γ^2 .

2

(ii) Find the value of $\alpha^3 + \beta^3 + \gamma^3$.

3

(b) Find a possible polynomial equation $P(x)$, of smallest degree that satisfies the following conditions:

3

- The polynomial $P(x)$ has rational coefficients.
- Two of its roots are $1 + \sqrt{5}$ and $-6i$.
- When $y = P(x)$ is drawn on a Cartesian plane, there are four x -intercepts.

(c) A complex number ω is such that $|\omega| = 1$.

2

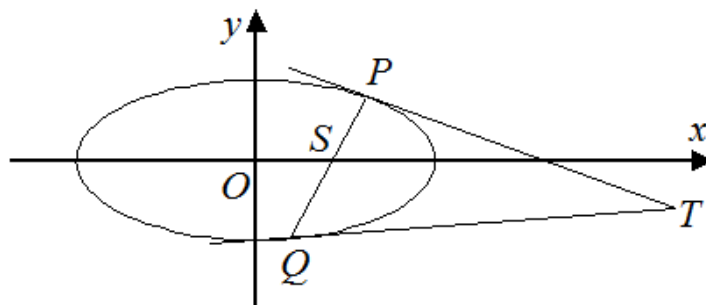
If $z = \frac{1+\omega}{1-\omega}$, find the locus of z as ω moves on the complex number plane.

(d) The point P lies on the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$, where $a > b$

The chord through P and the focus $S(ae, 0)$ meets the ellipse again at Q .

The tangents to the ellipse at P and Q meet at the point $T(x_0, y_0)$.

The equation of PQ is $\frac{xx_0}{a^2} + \frac{yy_0}{b^2} = 1$. **[DO NOT PROVE THIS]**



(i) Show that T lies on the directrix.

1

The point P is now chosen so that T lies on the x -axis.

(ii) What is the value of the ratio $PS:ST$?

1

(iii) Show that $\angle PTQ$ is acute.

1

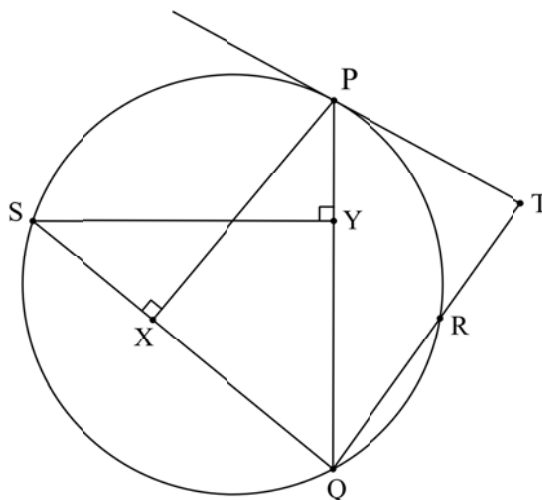
(iv) Show that the area of the triangle PQT is $b^2 \left(\frac{1}{e} - e \right)$.

2**End of Question 12**

Question 13 (15 marks)

Marks

- (a) In the diagram below, TP is the tangent of the circle at P , and TQ is a secant cutting the circle at R .
 SQ is a chord of the circle such PX and SY are perpendicular to SQ and PQ respectively.

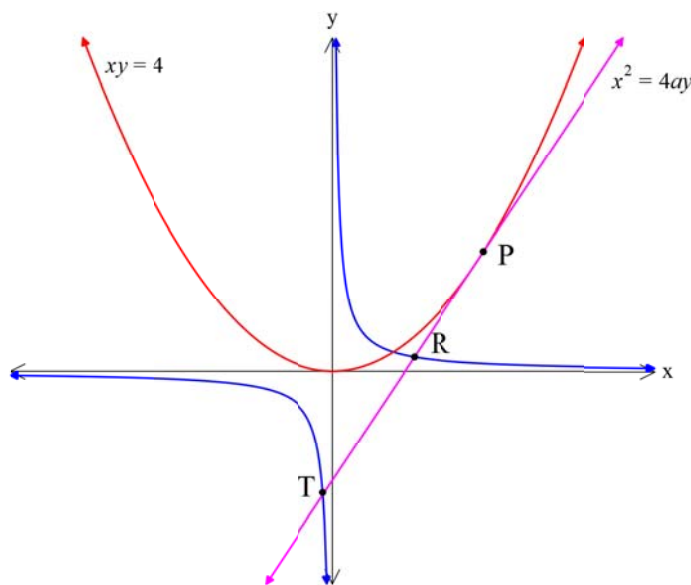


Copy the diagram in your answer booklet.

- | | |
|---|---|
| (i) Prove that $\angle TRP = \angle TPQ$. | 2 |
| (ii) Explain why $SPYX$ is a cyclic quadrilateral and state the diameter of the circle $SPYX$. | 2 |
| (iii) Prove $\angle PYX = \angle PRQ$. | 2 |

Question 13 continues on the next page

- (b) $P(8p, 4p^2)$ is a point on the parabola $x^2 = 16y$.
The tangent to the parabola at P cuts the hyperbola $xy = 4$ at two distinct points R and T .



- (i) M is the midpoint of RT , show that M has coordinates $(2p, -2p^2)$. 2
You may assume the tangent to the parabola at P is
 $y = px - 4p^2$. **[DO NOT PROVE THIS]**
- (ii) Find the equation of the locus of M , as P moves on the parabola $x^2 = 16y$, stating any restrictions. 3

- (c) Let $z = x + iy$ be any non-zero complex number.

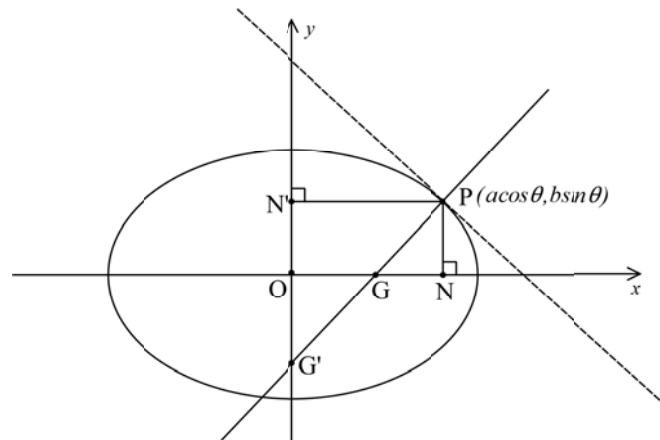
- (i) Show $z + \frac{1}{z} = x + \frac{x}{x^2+y^2} + i\left(y - \frac{y}{x^2+y^2}\right)$. 1
- (ii) Given that $z + \frac{1}{z} = k$, where k is real, show that $y = 0$ or $x^2 + y^2 = 1$. 1
- (iii) Show also that if $y = 0$ then $|k| \geq 2$ and that if $x^2 + y^2 = 1$ then $|k| \leq 2$. 2

End of Question 13

Question 14 (15 marks)

Marks

- (a) The diagram below shows the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$, where $a > b$.
The point $P(a \cos \theta, b \sin \theta)$ lies on the ellipse.



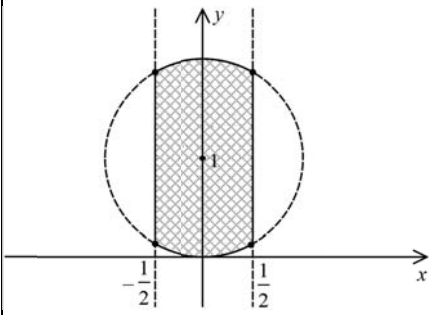
The normal to the ellipse at P meets the major and minor axes of the ellipse at G and G' respectively. N and N' are the feet of the perpendiculars from P to the major axes and minor axes respectively.

- (i) Show that the equation of the normal at P is $ax \sec \theta - by \operatorname{cosec} \theta = a^2 - b^2$. 2
- (ii) Show that the ratio of $OG : ON = e^2 : 1$. 2
- (iii) Given that $\Delta PN'G' \sim \Delta PNG$, find the ratio of the area of $\Delta PN'G' : \Delta PNG$. 1
- (b) (i) Prove $4 \cos \frac{\theta}{2} \cos \frac{\theta}{4} \sin \frac{\theta}{4} = \sin \theta$ 1
- (ii) Prove by Mathematical Induction that for any positive integer n 3
- $$\sin \theta = 2^n \cos \frac{\theta}{2} \cos \frac{\theta}{4} \dots \cos \frac{\theta}{2^n} \sin \frac{\theta}{2^n}$$
- (iii) Explain why 1
- $$\lim_{n \rightarrow \infty} \left(\frac{\sin \frac{\theta}{2^n}}{\frac{\theta}{2^n}} \times 2^n \right) = 1$$
- (iv) Hence using (ii), (iii) and choosing a suitable value for θ , show: 3
- $$\frac{2}{\pi} = \cos \frac{\pi}{4} \times \cos \frac{\pi}{8} \times \cos \frac{\pi}{16} \times \dots$$
- (v) Show that $\pi = 2 \left(\frac{2}{\sqrt{2}} \times \frac{2}{\sqrt{2+\sqrt{2}}} \times \dots \right)$ 2

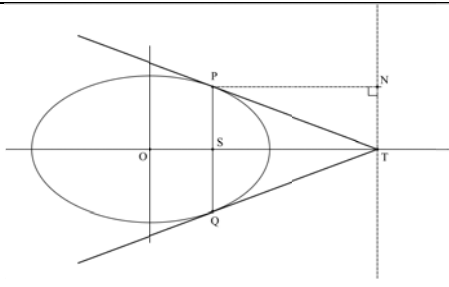
End of Exam

HSC Year12 Extension 2 Half Yearly 2017

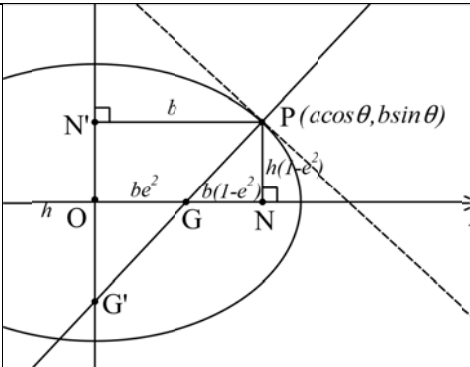
	Solutions	Mks	Comments
1	$b^2 = a^2(1 - e^2)$ $16 = 25(e^2 - 1)$ $e = \frac{3}{5}$	1	B
2	$w^4 = \sqrt[4]{3} \operatorname{cis}(4 \times \pi)$ $w^4 = 9 \operatorname{cis} 4\pi$ $w^4 = 9 \operatorname{cis} 0$ $w^4 = 9$	1	B
3	Let the roots be $\alpha, \beta, -1$ Product of roots = $-\frac{b}{a}$ $\alpha \times \beta \times -1 = -1$ $\beta = \frac{1}{\alpha}$	1	A
4	Let $P(x) = 3x^5 - 5x^4 + 5x - 3$ $P(1) = 0$ $P'(x) = 15x^4 - 20x^3 + 5$ $P'(1) = 0$ $P''(x) = 60x^3 - 60x^2$ $P''(1) = 5$ $P'''(x) = 180x^2 - 120x$ $P'''(1) = 60$ $\therefore P(x)$ has multiplicity 3	1	C
5	Triangular inequality	1	D
6	The roots are $2i, -2i, 2 + i, 2 - i$ $P(x) = (x^2 + 4)(x^2 - 4x + 5)$ $P(x) = x^4 - 4x^3 + 9x^2 - 16x + 20$ $a + b + c + d = -4 + 9 - 16 + 20$ $= 9$	1	C
7	$m_{OP} = \frac{b \tan \theta}{a \sec \theta}$ $m_{OQ} = \frac{b \tan \alpha}{a \sec \alpha}$ $m_{OP} \times m_{OQ} = -1$ $\frac{b \tan \theta}{a \sec \theta} \times \frac{b \tan \alpha}{a \sec \alpha} = -1$ $\frac{\cos^2 \theta}{\cos^2 \theta} \times \frac{b^2 \tan \theta}{a^2 \sec \theta} \times \frac{\tan \alpha}{\sec \alpha} = -1$ $\sin \theta \sin \alpha = -\frac{a^2}{b^2}$	1	C
8	Derive both side implicitly: $2x + \left(3 \times y + \frac{dy}{dx} \times 3x\right) = 10y \times \frac{dy}{dx}$ $2x + 3y = \frac{dy}{dx}(10y - 3x)$ $\frac{dy}{dx} = \frac{2x + 3y}{10y - 3x}$	1	A
9	If two particular people are together, then there are 6 entities, ie (AB)CDEFG. (AB) will have 4 rooms to choose from C will have 4 rooms to choose from ... G will have 4 rooms to choose from $\therefore$ there are 4^6 ways to place 6 entities.	1	D

	Solutions	Mks	Comments
10	(A) False as x is positive, and there are x -ordinates that are negative (B) False as $t \rightarrow \infty, x \rightarrow -\infty$ (C) False as x is positive, and there are x -ordinates that are negative (D) True as $-1 \leq \sin t \leq 1, -c \leq c \sin t \leq c$, ie $-c \leq x$ - ordinate $\leq c$	1	D
11a(i)	$3(2+i) + i(i-1) = 6 + 3i - 1 - i$ $= 5 + 2i$	1	1 mark • correct answer.
11a(ii)	$(2+i)(-1-i) = -1 - 3i$	2	2 mark • correct solution. 1 mark • correct conjugate • correct multiplication with a wrong expression of the conjugate.
11b(i)	$z = 2 \left[\cos\left(\frac{\pi}{6}\right) + i \sin\left(\frac{\pi}{6}\right) \right]$	1	1 mark • correct answer.
11b(ii)	$z^7 - 64z = z(z^6 - 64)$ $z^6 = 2^6 \left(\cos\frac{6\pi}{6} + i \sin\frac{6\pi}{6} \right)$ $z^6 = 64$ $z^7 - 64z = z(64 - 64)$ $= 0$	2	2 mark • correct solution. 1 mark • correct use of DM's Thm
11c(i)	$9 = -16(1 - e^2)$ $e = \frac{5}{4}$	1	1 mark • correct answer
11c(ii)	$\therefore \text{foci are } (\pm 5, 0)$ $x - \text{int when } y = 0$ $\frac{x^2}{16} = 1$ $x\text{-intercepts are } (\pm 4, 0)$	2	2 mark • correct solution. 1 mark • correct foci • correct intercept
11c(iii)	Directrices: $x = \pm \frac{16}{5}$ Eqn of asymptote: $y = \pm \frac{3}{4}x$	2	2 mark • correct solution. 1 mark • correct directrices • correct asymptote
11d	$ z + \bar{z} \leq 1$ $ 2x \leq 1$ $-\frac{1}{2} \leq x \leq \frac{1}{2}$ 	2	2 mark • correct solution. 1 mark • Shows region inside circle at centre (0,1) • Shows region between the lines $-\frac{1}{2} \leq x \leq \frac{1}{2}$ • Correct boundaries with incorrect region.
11e(i)	$\omega^3 - 64 = 0$ $(\omega - 4)(\omega^2 + 4\omega + 16) = 0$ Since ω is a complex root so $\omega \neq 4$ $\omega^2 + 4\omega + 16 = 0$ $\omega^2 = -4\omega - 16$ $\omega^2 = -4(\omega + 4)$	1	1 mark • correct solution.

	Solutions	Mks	Comments
11e(ii)	$(\omega + 4)^3 = \left(\frac{\omega^2}{-4}\right)^3$ $= \frac{(\omega^3)^2}{-64}$ $= \frac{(64)^2}{-64}$ $= -64$	1	1 mark • correct solution.
12a(i)	Let $A = x^2$, where $x = \alpha, \beta$ and γ $x = \sqrt{A}$ $\sqrt{A}^3 + 6\sqrt{A}^2 + 5\sqrt{A} - 2 = 0$ $A\sqrt{A} + 5\sqrt{A} = 2 - 6A$ Square both sides $A(A + 5)^2 = (2 - 6A)^2$ $A^3 - 26A^2 + 49A - 4 = 0$ $\therefore x^3 - 26x^2 + 49x - 4 = 0$	2	2 mark • correct solution. 1 mark • substitutes $x\sqrt{A}$
12a(ii)	$\alpha^3 + 6\alpha^2 + 5\alpha - 2 = 0$ $\beta^3 + 6\beta^2 + 5\beta - 2 = 0$ $\gamma^3 + 6\gamma^2 + 5\gamma - 2 = 0$ $\alpha^3 + \beta^3 + \gamma^3 + 6(\alpha^2 + \beta^2 + \gamma^2) + 5(\alpha + \beta + \gamma) - 2 \times 3 = 0$ $\alpha^3 + \beta^3 + \gamma^3 + 6(26) + 5(-6) - 2 \times 3 = 0$ $\alpha^3 + \beta^3 + \gamma^3 = -120$	3	3 mark • correct solution. 2 mark • significant progress to finding $\alpha^3 + \beta^3 + \gamma^3$ 1 mark • Finds $\alpha^2 + \beta^2 + \gamma^2$ and $\alpha + \beta + \gamma$ • Forms a polynomial with α^2, β^2 and γ^2
12b	If the polynomial has rational coefficients, then the roots must be $-6i, 6i, 1 + \sqrt{5}, 1 - \sqrt{5}$ As the equation $(x^2 + 36)(x^2 - 2x - 4)$ has two x-intercepts, it needs any two factors to have 4 x intercepts. The smallest degree must be 6. A possible equation is $x(x - 1)(x^2 + 36)(x^2 - 2x - 4)$	3	3 mark • correct solution. 2 mark • Two conditions met. 1 mark • One condition met
12c	$z - z\omega = 1 + \omega$ $z - 1 = \omega(1 + z)$ $ z - 1 = \omega \times 1 + z $ But $ \omega = 1$ $ z - 1 = z + 1 $ $\therefore \text{locus is } x = 0$	2	2 mark • correct solution. 1 mark • eliminates ω • simplifies z using $ \omega = 1$
12d (i)	PQ passes through $S(ae, 0)$ $\frac{ae x_0}{a^2} + \frac{0 \times y_0}{b^2} = 1$ $\therefore x_0 = \frac{a}{e}$ So $T(x_0, y_0)$ lies on the directrix $x = \frac{a}{e}$	1	1 mark • correct solution

	Solutions	Mks	Comments
12d(ii)	Since T is on the x-axis, We know that $\frac{PS}{PN} = e$ by the definition of the ellipse. Since $PSTN$ is a rectangle, $PN = ST$ $\therefore \frac{PS}{ST} = e$ 	1	1 mark • correct solution
12d(iii)	$\tan \angle PTS = \frac{PS}{PT}$ $= e < 1$ $\angle PTS < 45^\circ$ $\angle PTQ = 2\angle PTS < 90^\circ$	1	1 mark • correct solution
12d(iv)	Sub $x = ae$ into $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ $e^2 + \frac{y^2}{b^2} = 1$ $y^2 = b^2(1 - e^2)$ $PS^2 = b^2(1 - e^2)$ $\begin{aligned} \text{Area } \Delta PQT &= PS \times ST \\ &= PS \times \frac{PS}{e} \\ &= \frac{b^2(1 - e^2)}{e} \\ &= b^2 \left(\frac{1}{e} - e \right) \end{aligned}$	2	2 mark • correct solution. 1 mark • Finds PS^2 or PS • Correct area using the wrong value of PS .
13a(i)	Let $\angle TRP = \theta$ $\angle TRP = \angle PSQ = \theta$ (Exterior angle of cyclic quad = interior opposite angle) $\angle PSQ = \angle TPQ = \theta$ (alternate segment theorem) $\therefore \angle TPQ = \angle TRP = \theta$	2	2 marks • correct solution. 1 mark • significant progress towards a correct solution.
13a(ii)	$\angle SXP = \angle SYP = 90^\circ$ (angle in a semi circle) $SPYX$ is a cyclic quad with SP as a diameter.	2	2 marks • correct solution. 1 mark • Explains $SPYX$ is a cyclic quad • States SP is a diameter.
13a(iii)	$\angle PYX + \angle PSX = 180^\circ$ (opposite angles of a cyclic quad $SPYX$) $\angle PYX = 180^\circ - \theta$ $\angle PRQ + \angle PRT = 180^\circ$ (angles on a straight line) $\therefore \angle PRQ = 180^\circ - \theta$ $\angle PYX = \angle PRQ = 180^\circ - \theta$	2	2 mark • correct solution. 1 mark • significant progress towards solution
13b(i)	$y = px - 4p^2 \quad \text{--- (1)}$ $xy = 4 \quad \text{--- (2)}$ Let the intersection of the hyperbola and the parabola be $R(\alpha, y_R)$ and $T(\beta, y_T)$, $\begin{aligned} x(px - 4p^2) &= 4 \\ px^2 - 4p^2x - 4 &= 0 \\ \alpha + \beta &= \frac{4p^2}{p} \\ x_M &= \frac{4p}{2} \\ x_M &= 2p \end{aligned}$ $\begin{aligned} y_M &= p(2p) - 4p^2 \\ y_M &= -2p^2 \\ M &= (2p, -p^2) \end{aligned}$	2	2 marks • correct solution. 1 mark • finds x_M

	Solutions	Mks	Comments	
13b(iii)	Sub $p = \frac{x_M}{2}$ into y_M $y_M = -2 \times \left(\frac{x_M}{2}\right)^2$ $\therefore$ The locus of M lies on $y = -\frac{x^2}{2}$ At (0,0) the tangent at P does not touch the hyperbola. $\therefore x \neq 0$ Since P and Q is distinct, $P \neq Q$ $(2) \rightarrow y = \frac{4}{x}$ sub (2) $\rightarrow$ (1) $\frac{4}{x} = px - 4p^2$ $px^2 - 4p^2x - 4 = 0$ $\Delta > 0$ $16p^4 - 4 \times p \times -4 > 0$ $16p(p^3 + 1) > 0$ $16p \geq 0 \text{ or } p^3 < -1$ $4x > 0 \text{ or } \frac{x}{2} < -1$ $x > 0 \text{ or } x < -2$ <u>$\therefore x < -2 \text{ or } x > 0$</u>	3	3 marks • correct solution. 2 mark • finds any part of the correct restriction • correct inequality and solves. 1 mark • finds the equation of M lies on $y = -\frac{x^2}{2}$ • finds $16p(p^3 + 1) > 0$	
13c(i)	$z + \frac{1}{z} = x + iy + \frac{1}{x + iy}$ $= x + iy + \frac{x - iy}{x^2 + y^2}$ $= x + \frac{x}{x^2 + y^2} + i\left(y - \frac{y}{x^2 + y^2}\right)$	1	1 mark • correct solution	
13c(ii)	If k is real then $\operatorname{Im}\left(z + \frac{1}{z}\right) = 0$ $y - \frac{y}{x^2 + y^2} = 0$ $y(x^2 + y^2) - y = 0$ $y(x^2 + y^2 - 1) = 0$ $y = 0 \text{ or } x^2 + y^2 = 1$	1	1 mark • correct solution	
13c(iii)	If $y = 0$ $x + \frac{x}{x^2 + 0^2} = k$ $x + \frac{1}{x} = k$ $x^2 + 1 = kx$ $x^2 - kx + 1 = 0$ $\Delta \geq 0$ $k^2 - 4 \geq 0$ $k^2 \geq 4$ $ k \geq 2$	If $x^2 + y^2 = 1$ $x + x = k$ $x = \frac{k}{2}$ Since $x^2 + y^2 = 1$, The x-ordinate must be ≤ 1 $ x \leq 1$ $ k \leq 2$	2	2 mark • correct solution. 1 mark • finds one inequality for k

	Solutions	Mks	Comments	
14a(i)	Differentiating both sides: $\frac{2x}{a^2} + \frac{2y}{b^2} \times \frac{dy}{dx} = 0$ $\frac{dy}{dx} = -\frac{b^2 x}{a^2 y}$ $m_T = -\frac{b}{a} \cot \theta$ $m_N = \frac{a}{b} \tan \theta$ Equation of normal: $y - b \sin \theta = \frac{a}{b} \tan \theta (x - a \cos \theta)$ $by \cos \theta - b^2 \sin \theta \cos \theta = ax \sin \theta - a^2 \sin \theta \cos \theta$ Multiply both sides by $b \cos \theta$ $by \cos \theta - b^2 \sin \theta \cos \theta = ax \sin \theta - a^2 \sin \theta \cos \theta$ $\sin \theta \cos \theta (a^2 - b^2) = ax \sin \theta - by \cos \theta$ $a^2 - b^2 = ax \sec \theta - by \operatorname{cosec} \theta$	2	2 mark • correct solution. 1 mark • finds m_N	
14a(ii)	G occurs at the xint $a^2 - b^2 = ax \sec \theta$ $x = \frac{a^2 - b^2}{a \sec \theta}$ $OG = \frac{a^2 - b^2}{a \sec \theta}$ $ON = a \cos \theta \text{ (x - ordinate of P)}$ $\frac{OG}{ON} = \frac{a^2 - b^2}{a \sec \theta} \times \frac{1}{a \cos \theta}$ $\text{but } b^2 = a^2(1 - e^2)$ $b^2 - a^2 = a^2 e^2$ $\frac{OG}{ON} = \frac{a^2 e^2}{a^2}$ $\frac{OG}{ON} = e^2$ $OG:ON = e^2:1$	2	2 mark • correct solution. 1 mark • finds m_N	
14a(iii)	Since ΔPNG and $\Delta PN'G'$ are similar triangles and $OG:ON = e^2$ $OG = e^2 \times ON$ $OG = e^2 \times PN'$ $GN = PN'(1 - e^2)$ $G'N' = PN \times (1 - e^2)$ $\therefore \text{Area } \Delta PN'G':\text{Area } \Delta PNG = 1:(1 - e^2)^2$		1	1 mark • correct ratio.

	Solutions	Mks	Comments
14b(i)	$ \begin{aligned} LHS &= \cos \frac{\theta}{2} \cos \frac{\theta}{4} \sin \frac{\theta}{4} \\ &= \cos \frac{\theta}{2} \left(2 \times \cos \frac{\theta}{4} \sin \frac{\theta}{4} \right) \times \frac{1}{2} \\ &= \cos \frac{\theta}{2} \sin \frac{\theta}{2} \times 2 \times \frac{1}{2} \times \frac{1}{2} \\ &= \frac{1}{4} \sin \theta \\ &= RHS \end{aligned} $	1	1 mark • correct proof.
14b(ii)	Let $n = 1$ LHS = $\sin \theta$ $ \begin{aligned} RHS &= 2^1 \cos \frac{\theta}{2^1} \sin \frac{\theta}{2^1} \\ &= \sin \theta \quad (\text{double angle formula}) \end{aligned} $ Assume true for $n = k$ $\sin \theta = 2^k \cos \frac{\theta}{2} \cos \frac{\theta}{4} \dots \cos \frac{\theta}{2^k} \sin \frac{\theta}{2^k}$ Prove true for $n = k + 1$ AIM: $\sin \theta = 2^{k+1} \cos \frac{\theta}{2} \cos \frac{\theta}{4} \dots \cos \frac{\theta}{2^{k+1}} \sin \frac{\theta}{2^{k+1}}$ $ \begin{aligned} RHS &= 2^{k+1} \cos \frac{\theta}{2} \cos \frac{\theta}{4} \dots \cos \frac{\theta}{2^k} \cos \frac{\theta}{2^{k+1}} \sin \frac{\theta}{2^{k+1}} \\ &= 2^k \cos \frac{\theta}{2} \cos \frac{\theta}{4} \dots \cos \frac{\theta}{2^k} \times 2 \cos \frac{\theta}{2^{k+1}} \sin \frac{\theta}{2^{k+1}} \\ &= 2^k \cos \frac{\theta}{2} \cos \frac{\theta}{4} \dots \cos \frac{\theta}{2^k} \times \sin \frac{\theta}{2^k} \quad (\text{double angle formula}) \\ &= \sin \theta \quad (\text{if the assumption is true}) \\ &= LHS \end{aligned} $ ∴ If it is true for $n = k + 1$, if it is true for $n = k$ Since it is also true for $n = 1$, then it is true for positive integer of n by Mathematical Induction.	3	3 mark • correct solution. 2 mark • Completes all of the conditions below. 1 mark • Completes one of the conditions below. Conditions: (A) Tests for $n = 1$, and writes the correct assumption (B) Shows the use of the assumption in a correct expression of $n = k + 1$
14b(iii)	$\lim_{n \rightarrow \infty} \left(\frac{\sin \left(\frac{\theta}{2^n} \right)}{\frac{\theta}{2^n}} \times 2^n \right) = \lim_{n \rightarrow \infty} \left(\frac{\sin \left(\frac{\theta}{2^n} \right)}{\left(\frac{\theta}{2^n} \right)} \right)$ if $n \rightarrow \infty$, $\frac{\theta}{2^n} \rightarrow 0$, so $\frac{\theta}{2^n}$ is a very small angle $\lim_{n \rightarrow \infty} \left(\frac{\sin \left(\frac{\theta}{2^n} \right)}{\left(\frac{\theta}{2^n} \right)} \right) = \lim_{\left(\frac{\theta}{2^n} \right) \rightarrow 0} \left(\frac{\sin \left(\frac{\theta}{2^n} \right)}{\left(\frac{\theta}{2^n} \right)} \right) = 1$	1	1 mark • Correct proof and <u>must explain</u> $\frac{\theta}{2^n}$ is a small angle.

	Solutions	Mks	Comments
14b(iv)	Using the expression in part (ii), divide both side by θ $\frac{\sin \theta}{\theta} = 2^n \cos \frac{\theta}{2} \cos \frac{\theta}{4} \dots \cos \frac{\theta}{2^n} \frac{\sin \frac{\theta}{2^n}}{\frac{\theta}{2^n}}$ As $n \rightarrow \infty$, ans using (iii) $\lim_{n \rightarrow \infty} \left(\frac{\sin \theta}{\theta} \right) = \lim_{n \rightarrow \infty} \left(\cos \frac{\theta}{2} \cos \frac{\theta}{4} \dots \cos \frac{\theta}{2^n} \frac{\sin \frac{\theta}{2^n}}{\frac{\theta}{2^n}} \times 2^n \right)$ $\frac{\sin \theta}{\theta} = \cos \frac{\theta}{2} \cos \frac{\theta}{4} \dots \dots \times 1$ By substituting $\theta = \frac{\pi}{2}$ $\frac{\sin \frac{\pi}{2}}{\frac{\pi}{2}} = \cos \frac{\left(\frac{\pi}{2}\right)}{2} \cos \frac{\left(\frac{\pi}{2}\right)}{4} \cos \frac{\left(\frac{\pi}{2}\right)}{8} \dots$ $\frac{2}{\pi} = \cos \frac{\pi}{4} \cos \frac{\pi}{8} \cos \frac{\pi}{16} \dots$	3	3 mark • correct solution 2 mark • Completes two of the conditions below. 1 mark • Completes one of the conditions below. Conditions: (A) Uses $\theta = \frac{\pi}{2}$ (B) Manipulates expression to use the limit in (iii) (C) Uses (ii) and divides both sides by θ
14b(v)	$\cos \frac{\pi}{4} = \frac{\sqrt{2}}{2}$ we know: $\cos 2A = 2 \cos^2 A - 1$ $\cos A = \sqrt{\frac{\cos 2A + 1}{2}}$ $\cos \frac{\pi}{8} = \sqrt{\frac{\cos \frac{\pi}{4} + 1}{2}}$ $= \sqrt{\frac{\frac{\sqrt{2}}{2} + 1}{2}}$ $= \sqrt{\frac{\sqrt{2} + 2}{4}}$ $= \frac{\sqrt{\sqrt{2} + 2}}{2}$ $\cos \frac{\pi}{16} = \sqrt{\frac{\cos \frac{\pi}{8} + 1}{2}}$ $= \sqrt{\frac{\frac{\sqrt{\sqrt{2} + 2}}{2} + 1}{2}}$ $= \sqrt{\frac{\sqrt{\sqrt{2} + 2} + 2}{4}}$ $= \frac{\sqrt{\sqrt{\sqrt{2} + 2} + 2}}{2}$ $\therefore \frac{\pi}{2} = \frac{2}{\sqrt{2}} \times \frac{2}{\frac{\sqrt{\sqrt{2} + 2}}{2}} \times \frac{2}{\sqrt{\sqrt{\sqrt{2} + 2} + 2}} \dots \dots$ $\therefore \pi = 2 \left(\frac{2}{\sqrt{2}} \times \frac{2}{\frac{\sqrt{\sqrt{2} + 2}}{2}} \times \frac{2}{\sqrt{\sqrt{\sqrt{2} + 2} + 2}} \dots \dots \right)$	2	2 mark • correct solution, showing upto $\cos \frac{\pi}{8}$. 1 mark • Finds the exact value of $\cos \frac{\pi}{4}$ and attempt to use the double angle to find $\cos \frac{\pi}{8}$